

Smoke and flames rise from the site of airstrikes on an oil depot in Tehran on March 7, 2026. Sasan/Middle East Images

Aerosols in warfare

Aerosol Physics I
Miikka Silfverberg

How are aerosols related to war?

Used as weapons

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- **Incendiaries** are used to start **fires** and produce **smoke**. E.g. napalm and white phosphorous



AMNESTY INTERNATIONAL

31 October 2023

Lebanon: Evidence of Israel's unlawful use of white phosphorus in southern Lebanon as cross-border hostilities escalate

https://www.amnesty.org/en/latest/news/2023/10/lebanon-evidence-of-israels-unlawful-use-of-white-phosphorus-in-southern-lebanon-as-cross-border-hostilities-escalate/

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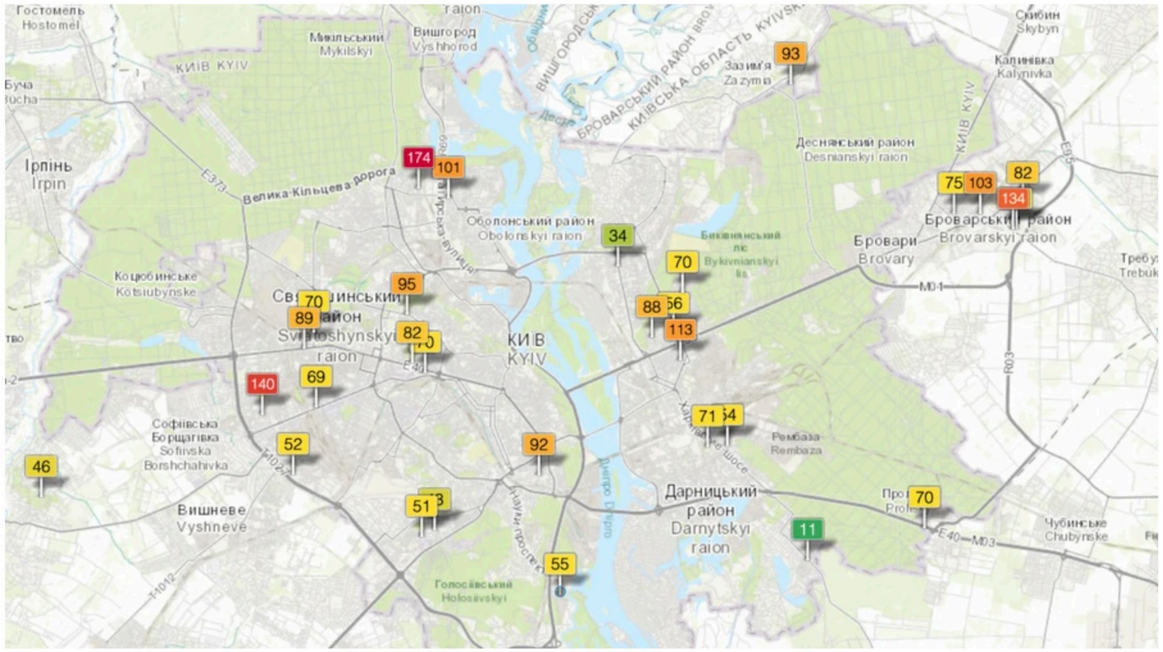
Byproducts of warfare

- Airstrikes can generate dust and smoke from fires. This can become a major problem in urban warfare

UKRAINSKA PRAVDA

Air quality in Kyiv deteriorates sharply after Russian attack sparks major fires

IVAN DIAKONOV — 5 AUGUST, 05:47



AIR QUALITY. MAP: AQICN

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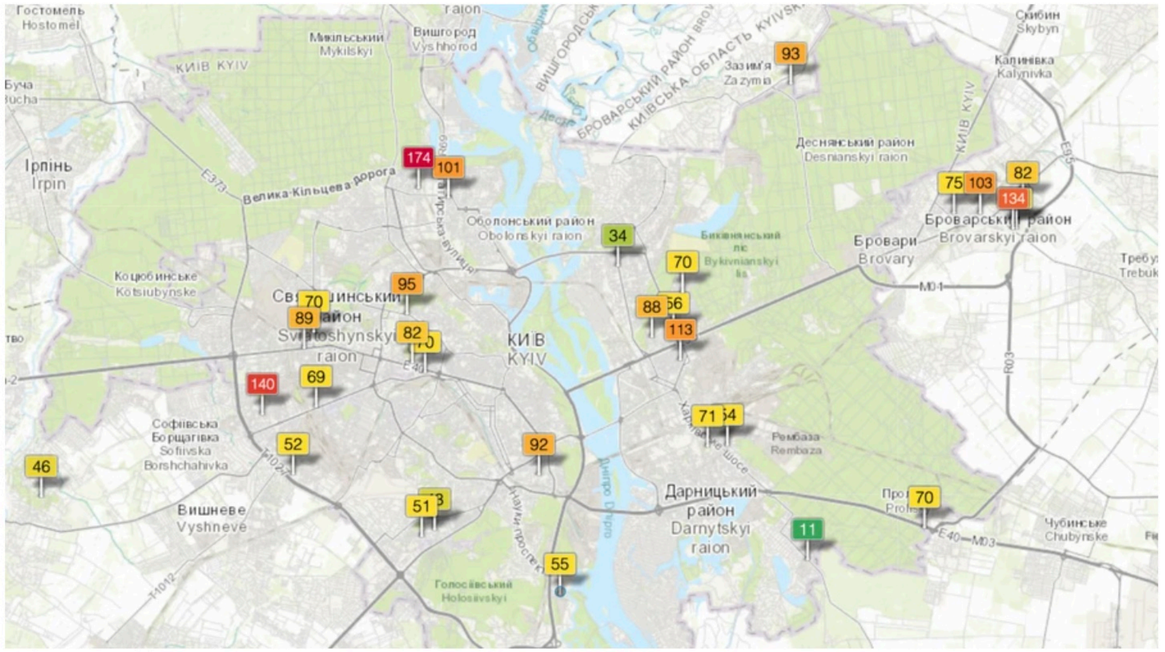
Byproducts of warfare

- Airstrikes can generate dust and smoke from fires. This can become a major problem in urban warfare
- Strikes on oil infrastructure release large quantities of primary aerosols like soot and aerosol precursors like SO2 and other gasses into the atmosphere

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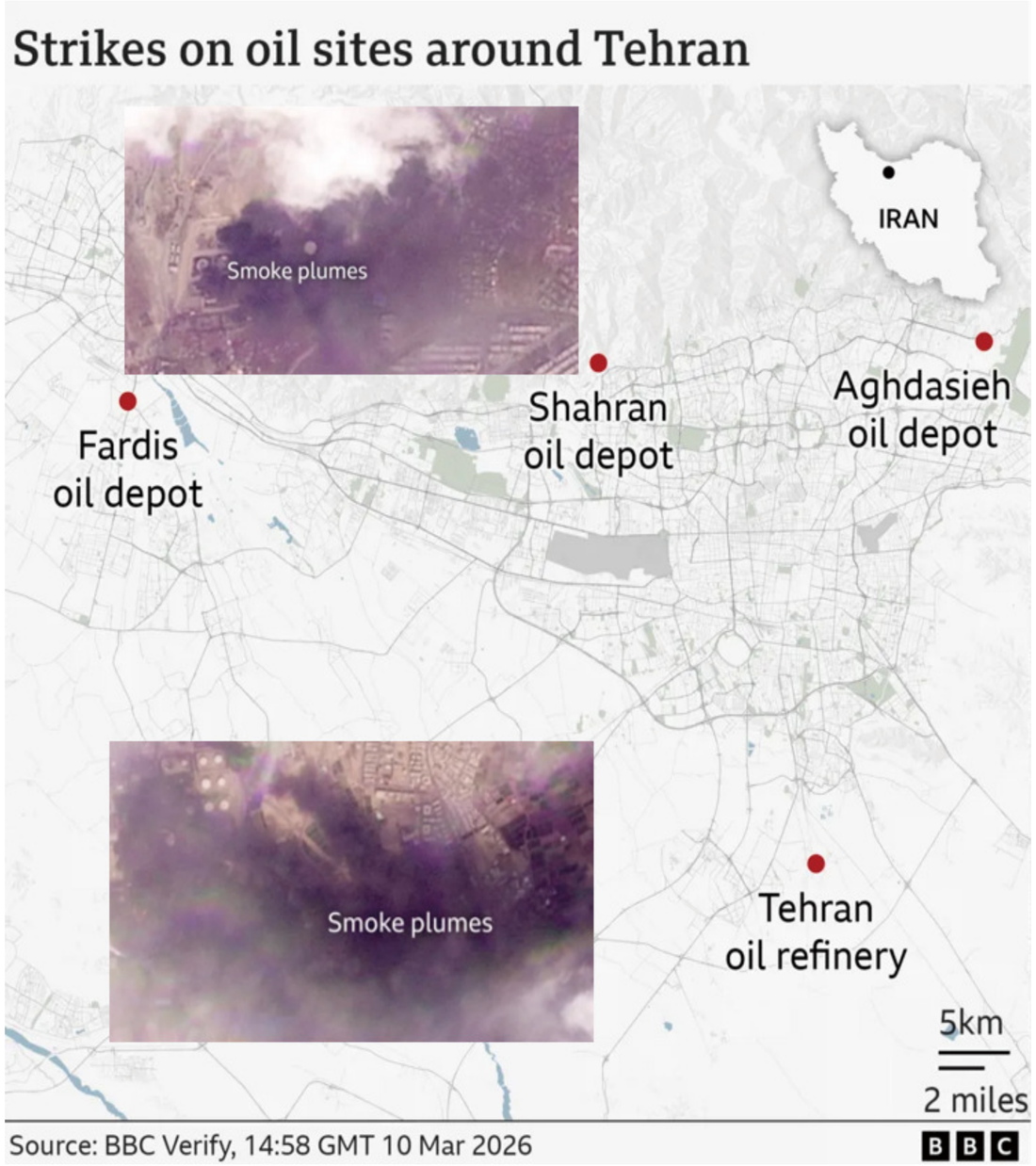


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March 7 strikes on Iran

- Airstrikes on oil refineries and depots triggered explosions and fires (Yin et al., 2026)

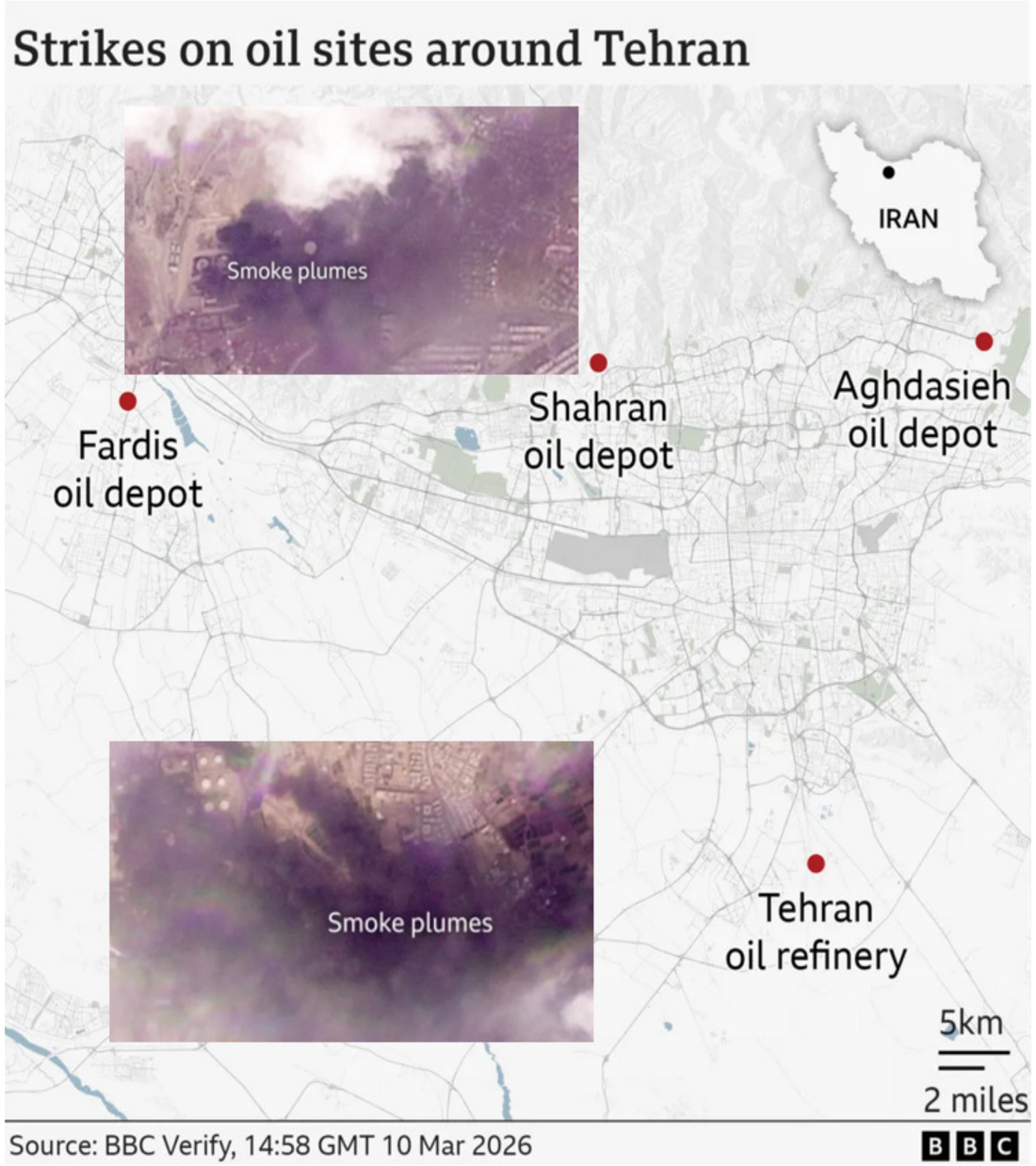


Four oil storage and refining facilities were targeted by U.S.-Israeli airstrikes. Satellite images of black smoke from the fires are shown from the Shahran oil depot (top) and Tehran oil refinery (bottom).

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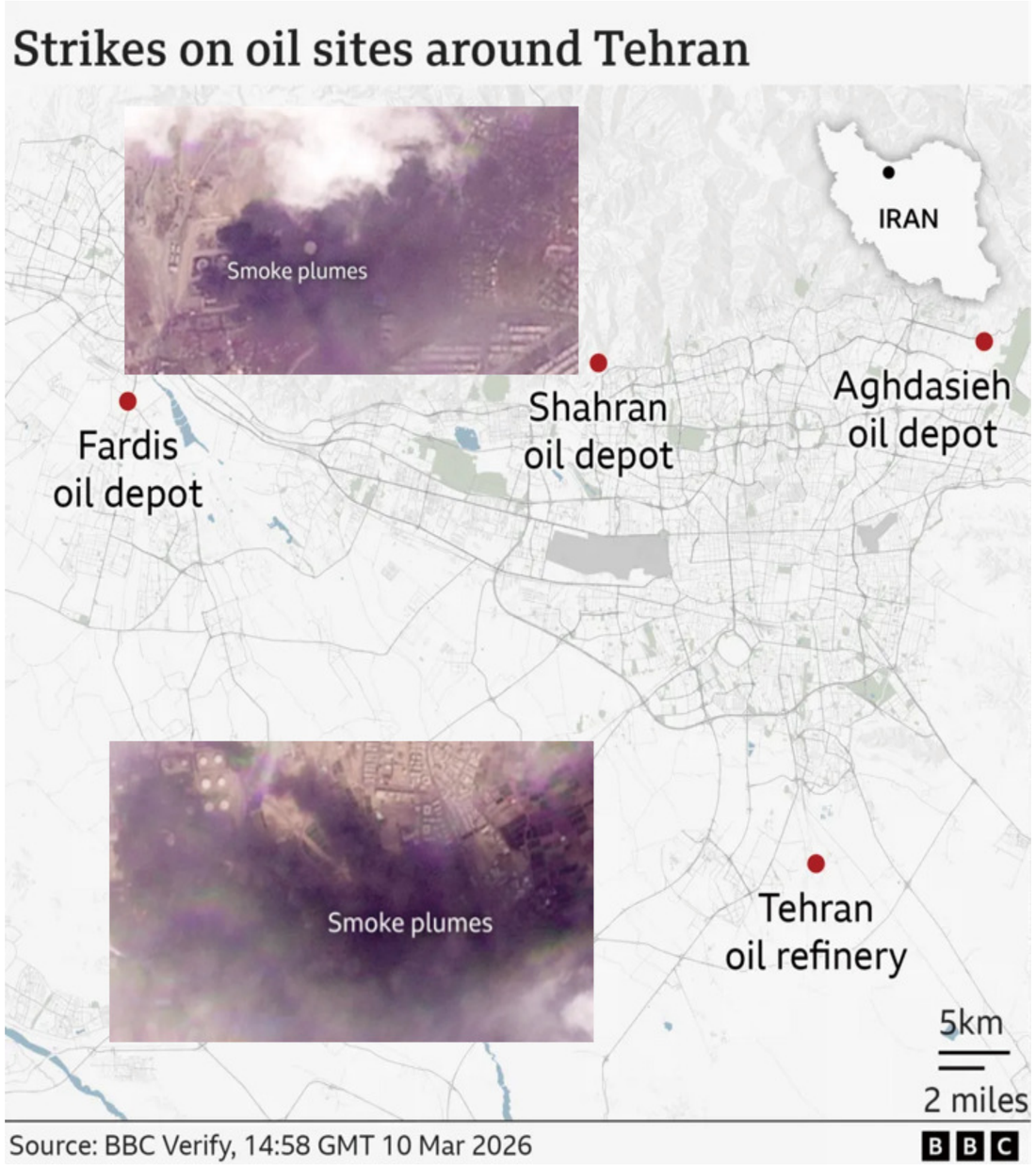


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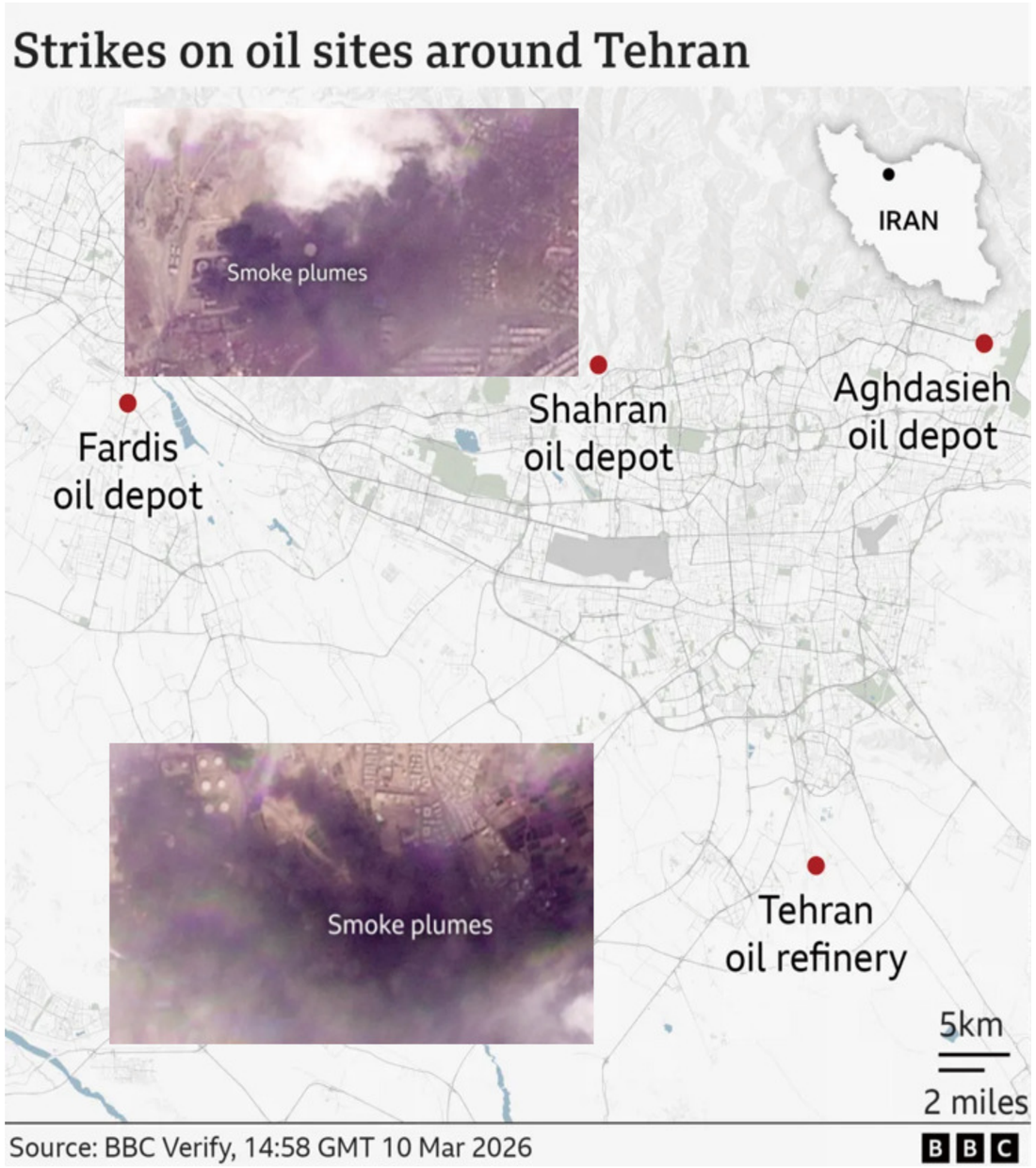


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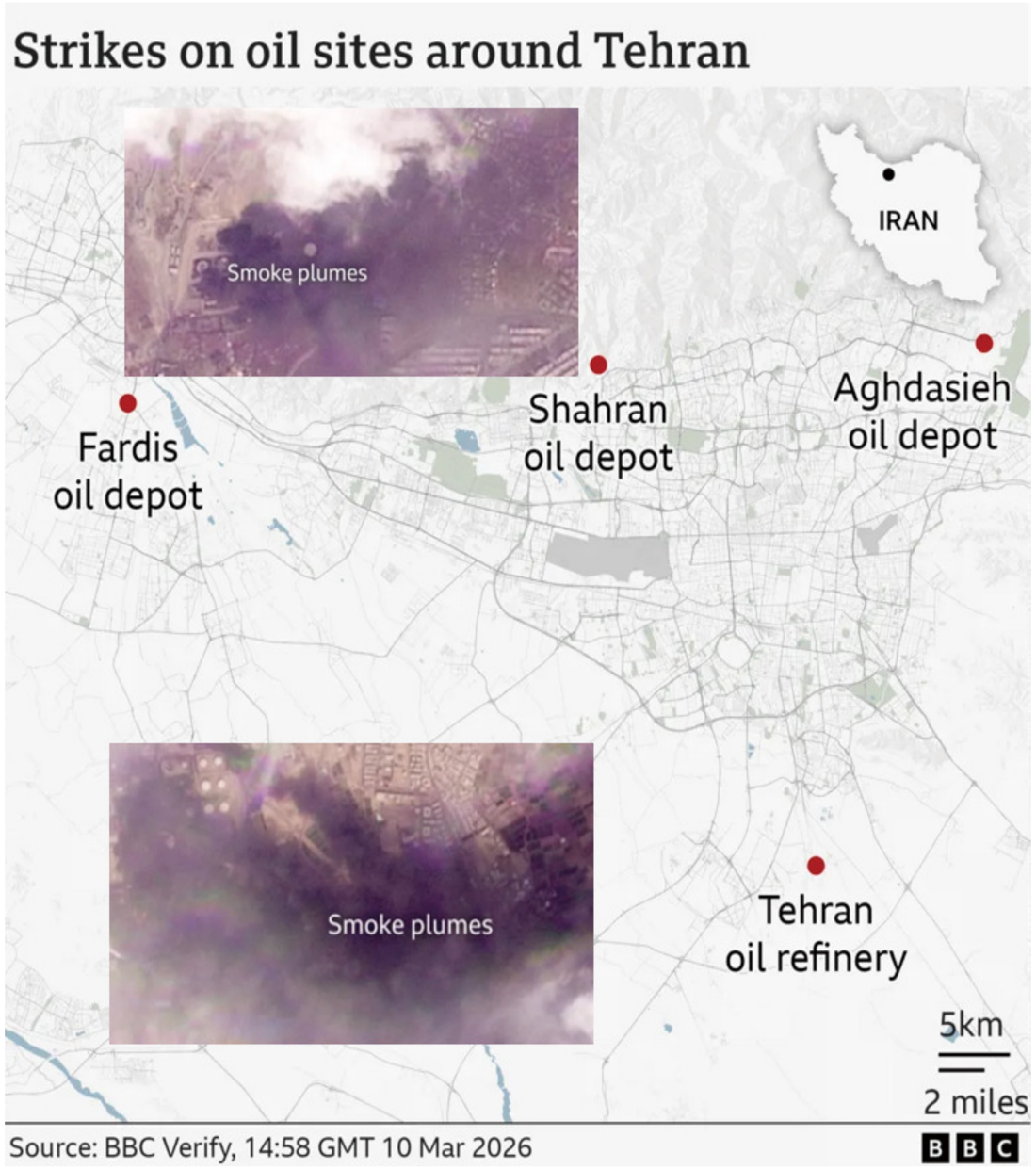


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- Yin et al., (2026) analyze the evolution of the SO2 emission cloud from the Tehran strikes



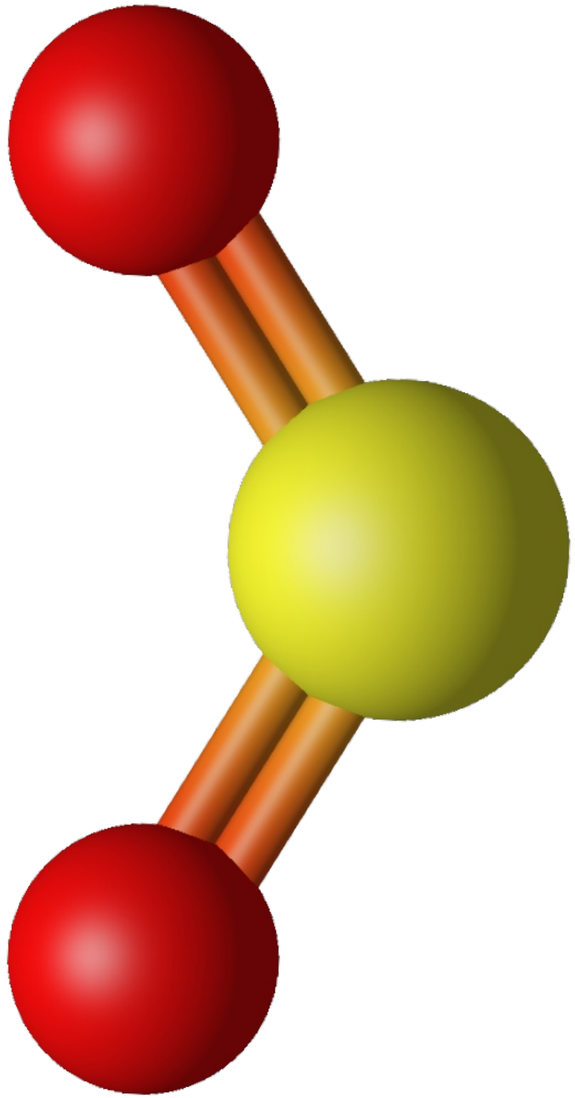
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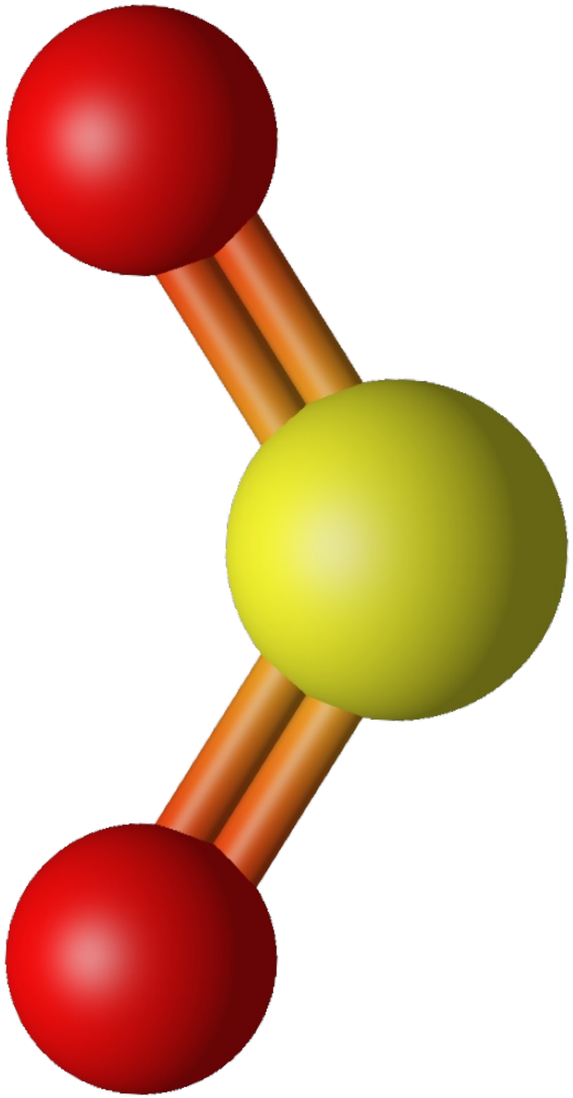


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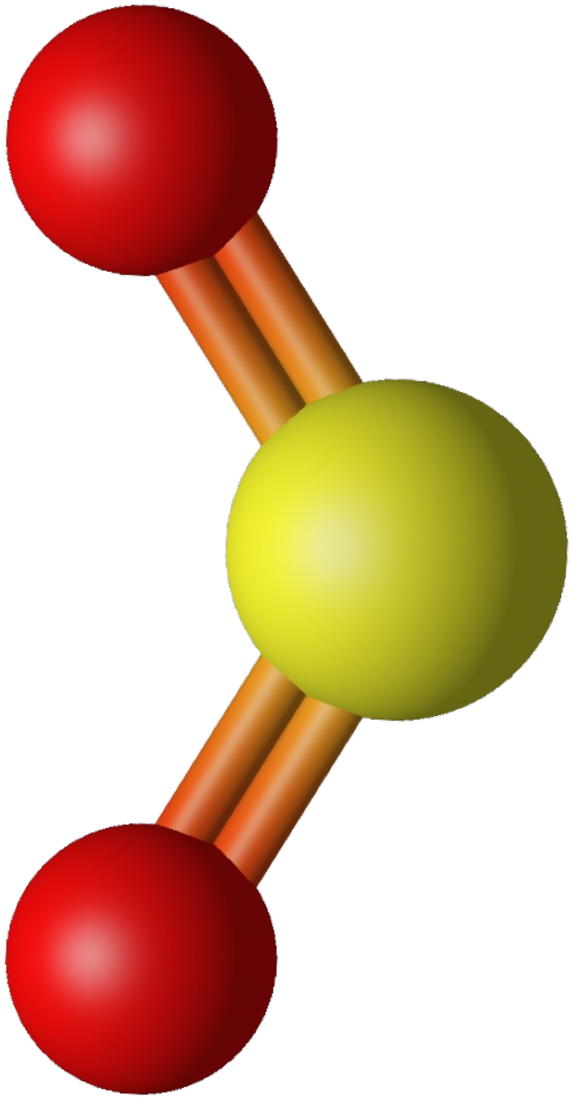


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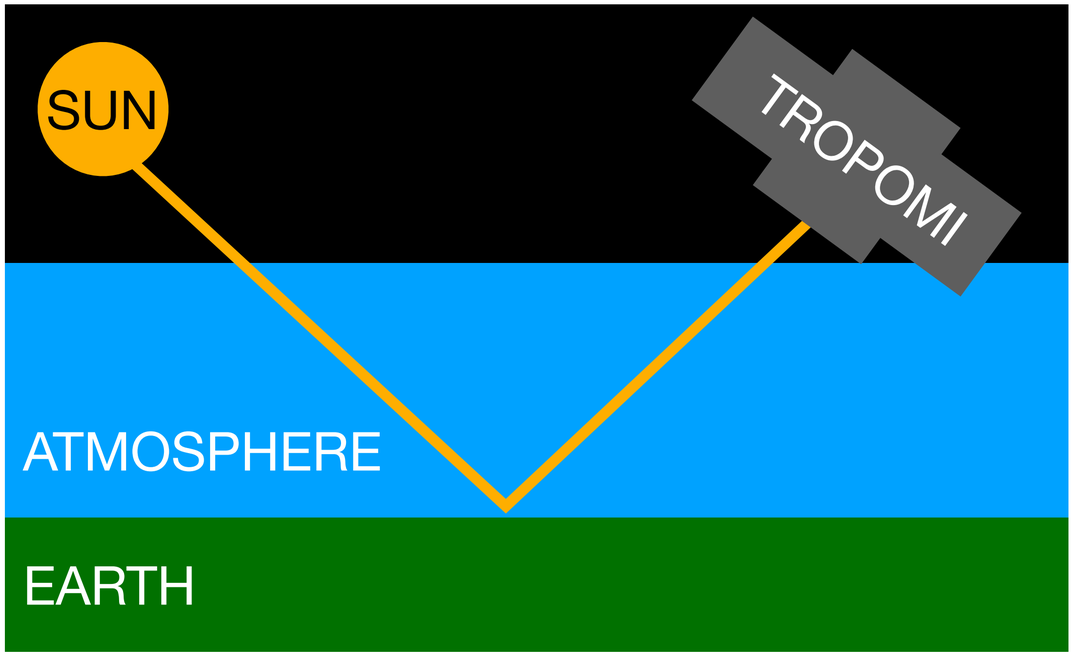
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- High concentrations of SO2 and secondary aerosol particles have health effects, particularly for vulnerable populations like people with respiratory illnesses, particularly children (EPA)



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The TROPOMI instrument

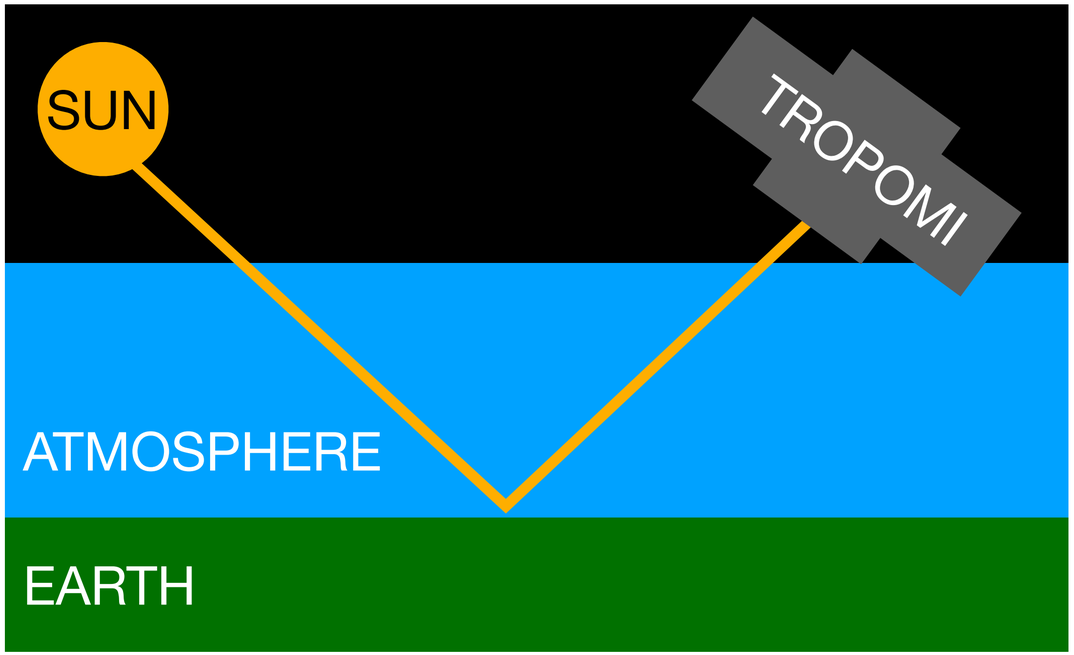
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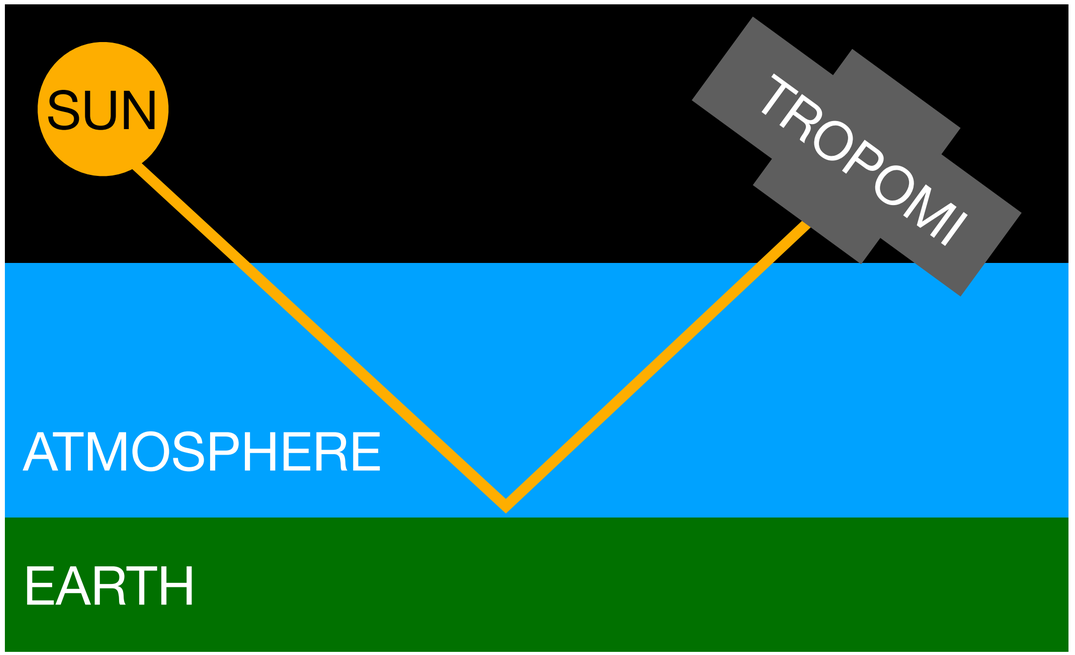
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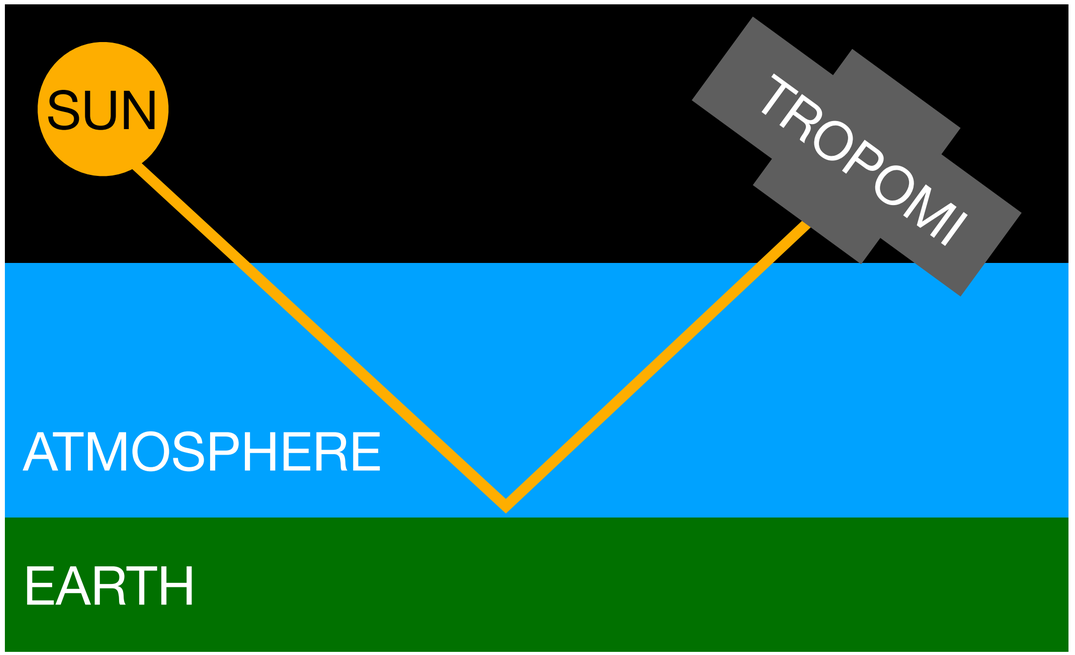
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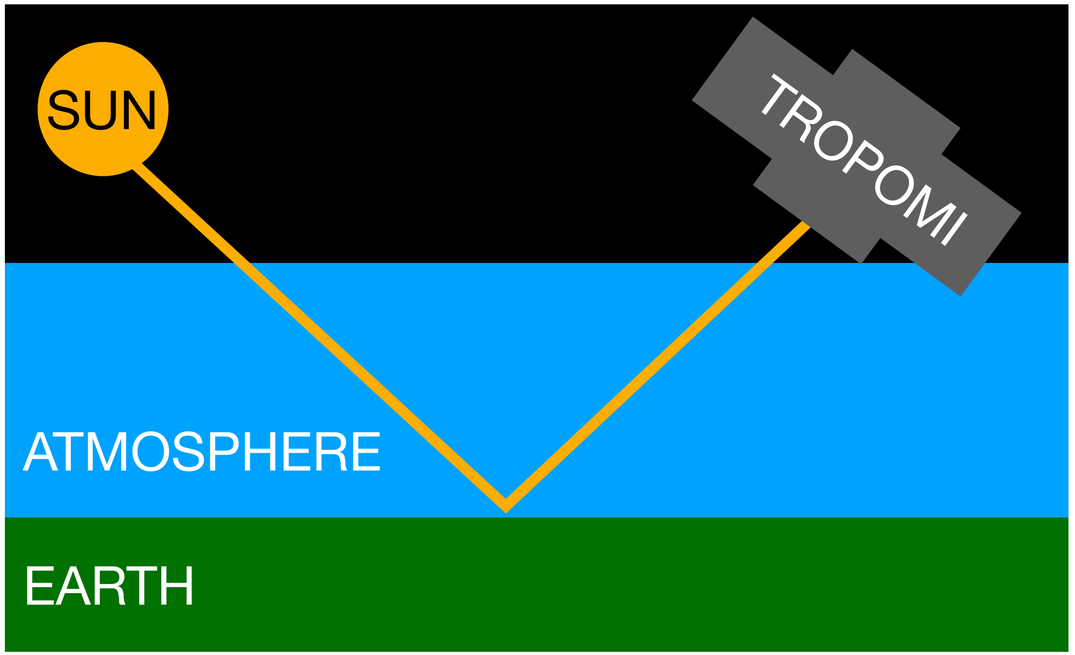
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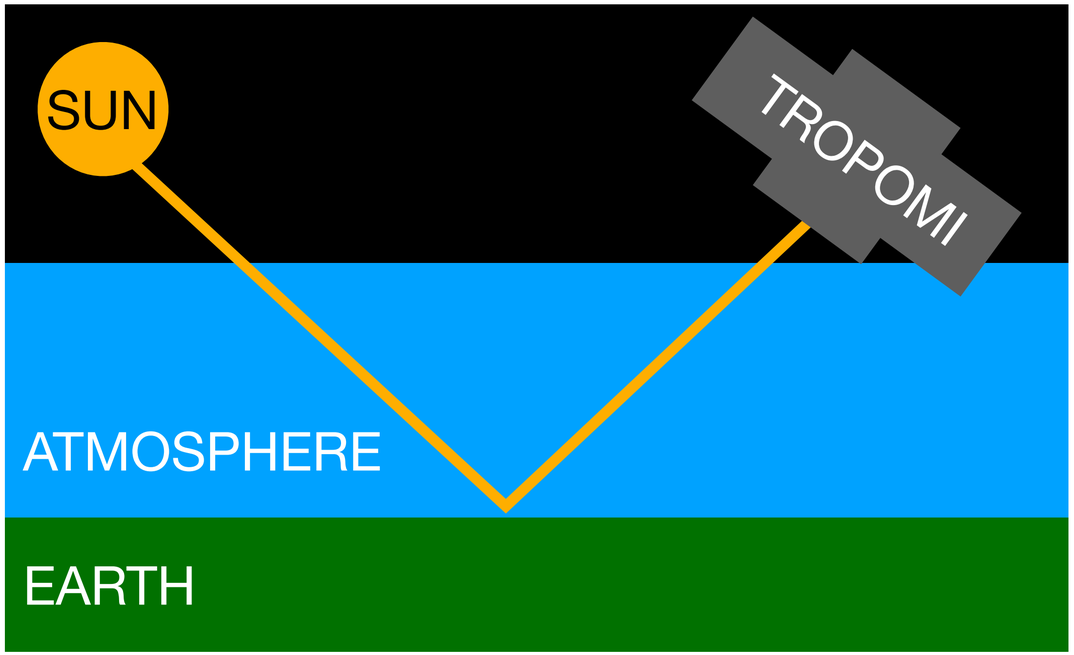


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- The **Lambert-Beer law** (Hinds, 1999, p 352) describes how the spectrum changes when passing through the atmosphere:

$$I(\lambda) = I_0(\lambda)e^{-\sigma_e(\lambda)L}$$



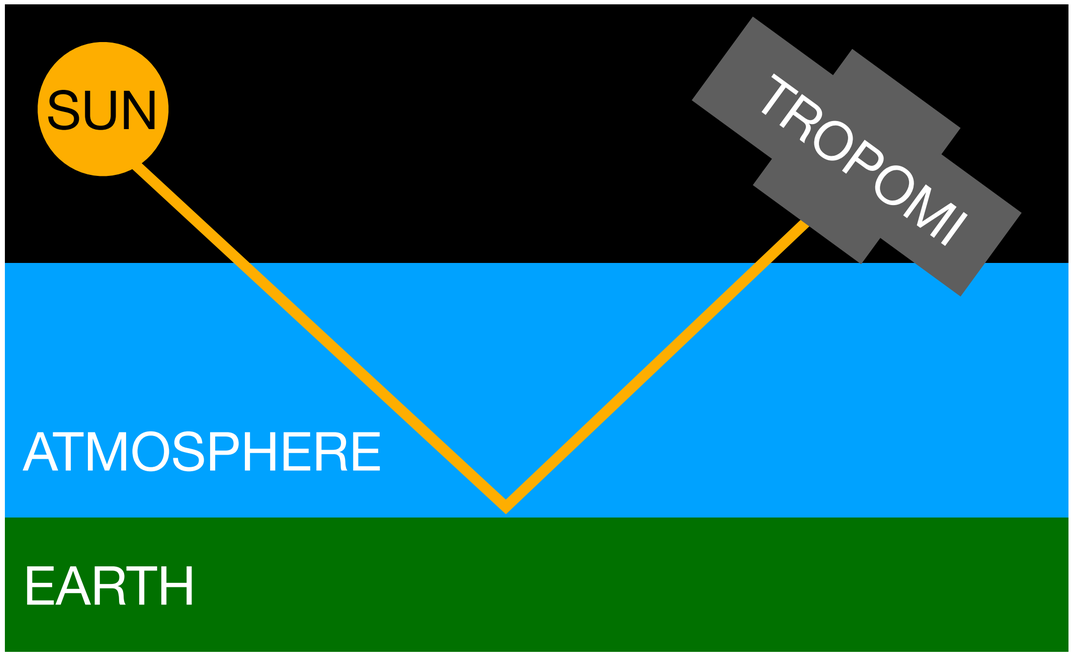
L – Length of idealized optical path in the atmosphere

σ_e – Wavelength-dependent extinction coefficient

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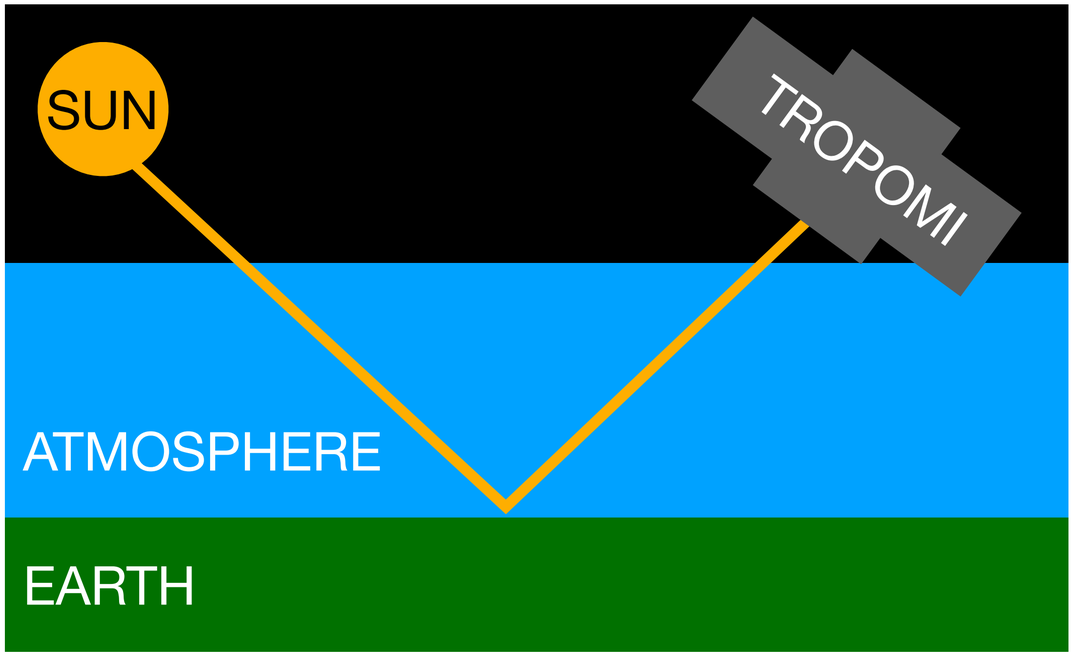
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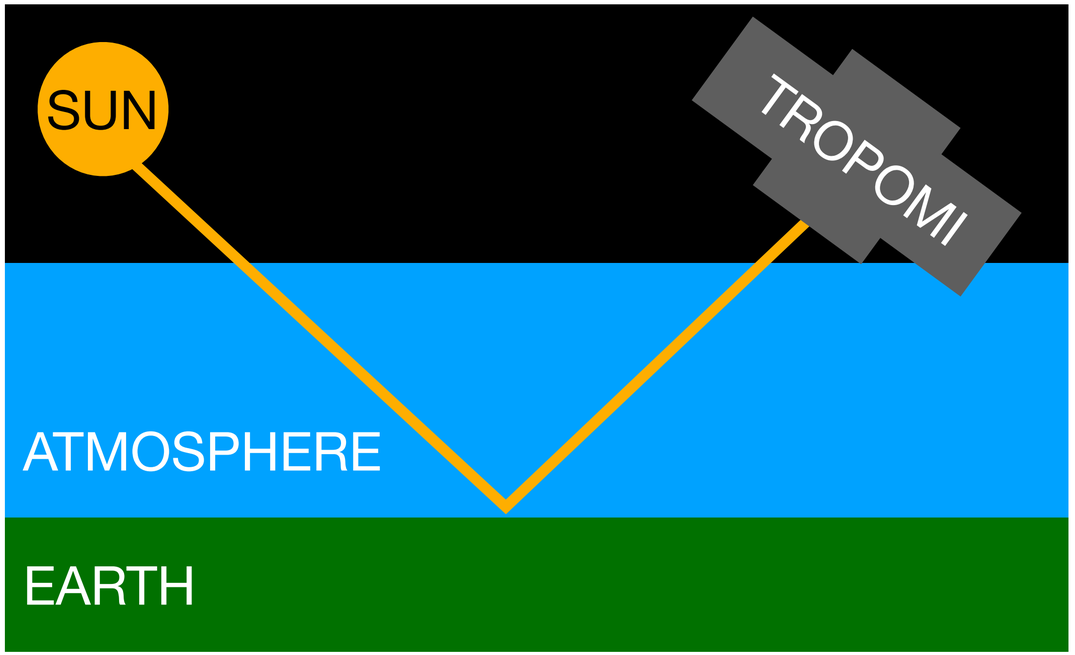
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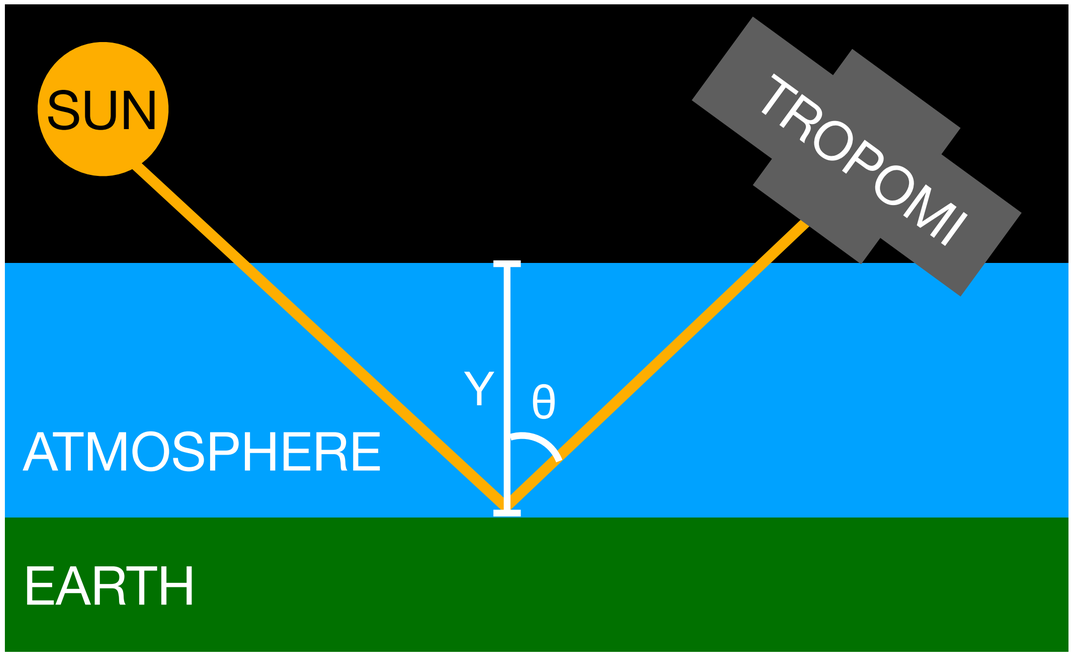
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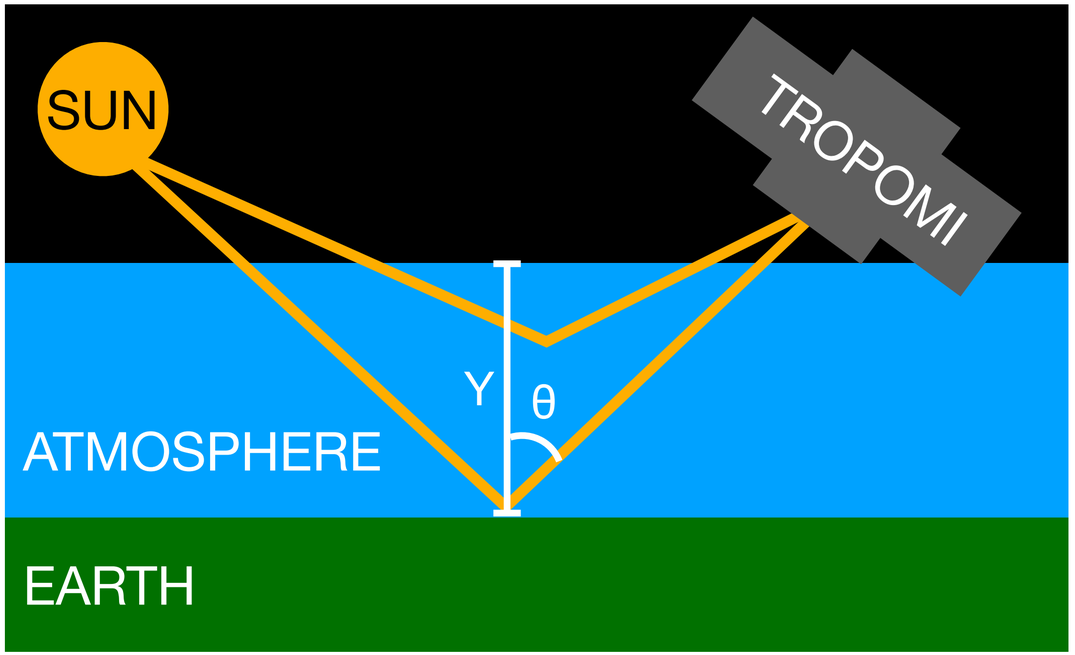
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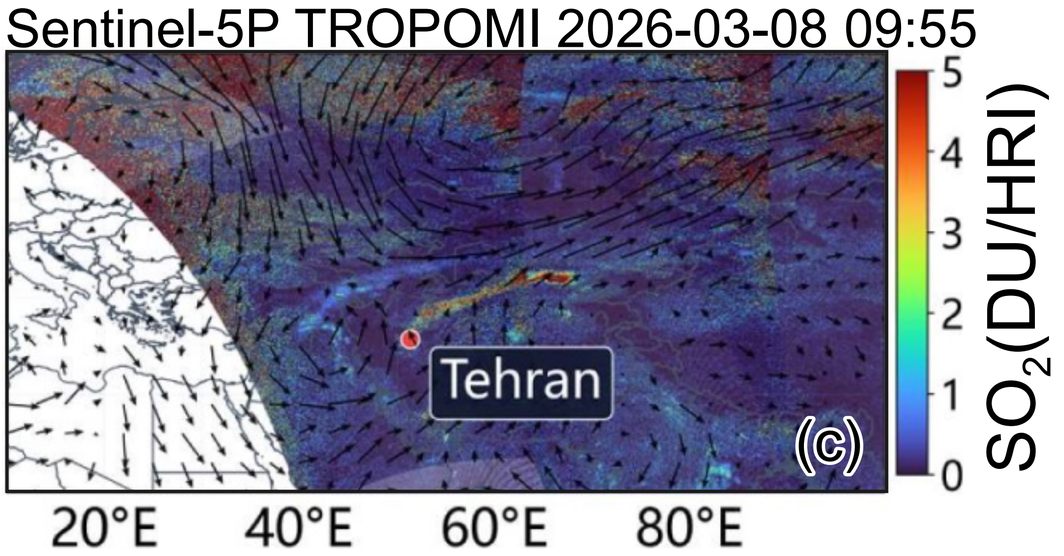
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- In reality, the corrections are more complex because light is scattered inside the atmosphere so there are many path lengths L



Yin et al. (2026)

Analysis of SO2 Emissions and Dispersion from the Fire at the Oil Storage and Refining Plant in Tehran, Iran, in March 2026

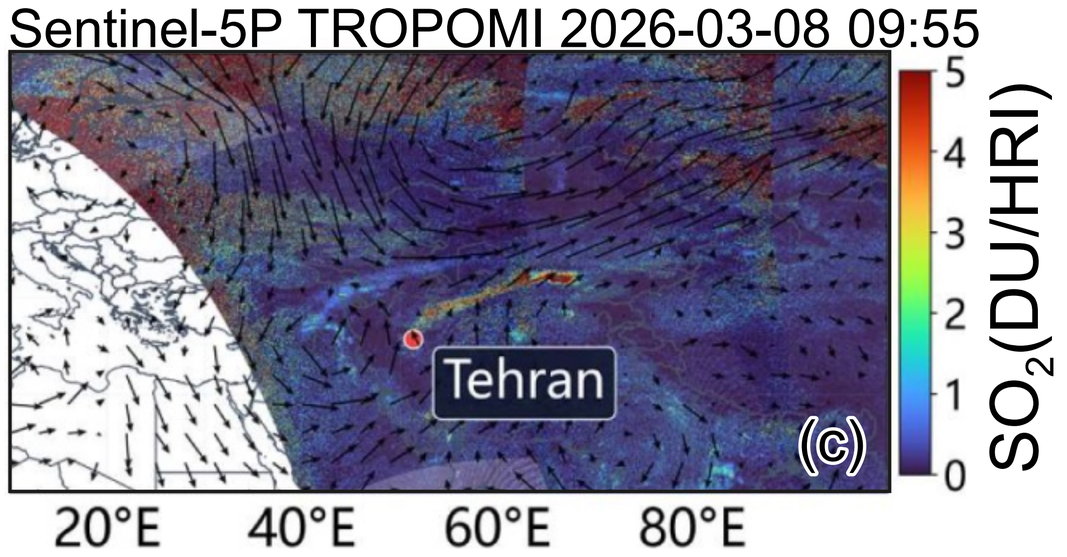
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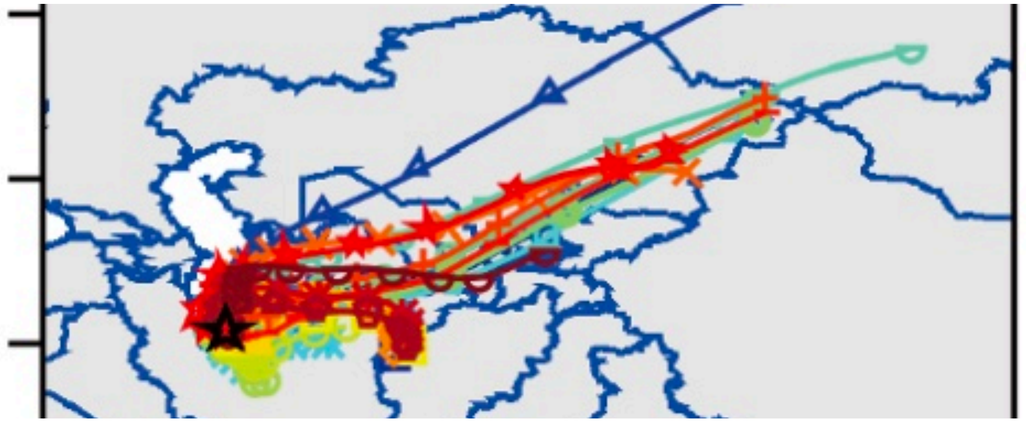


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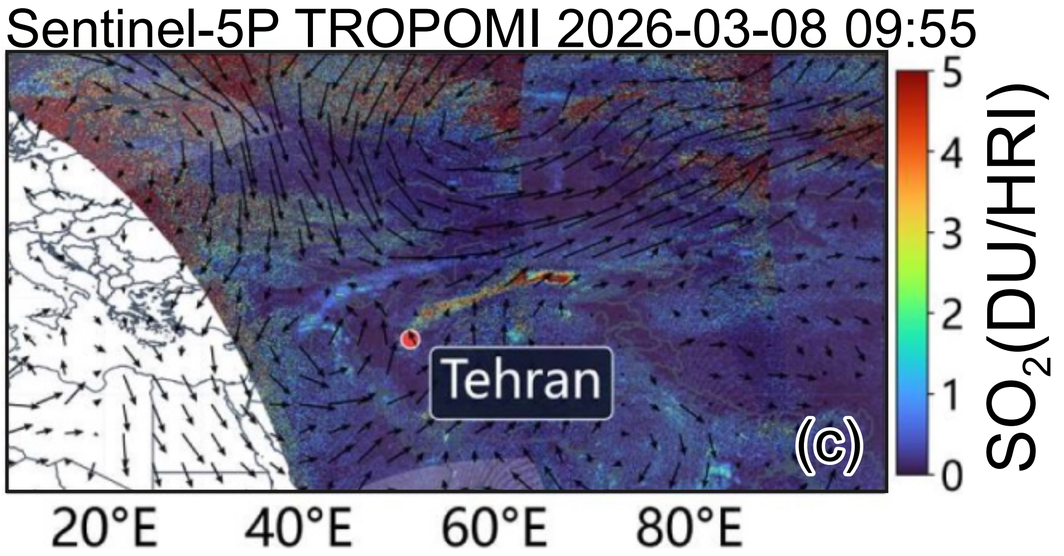


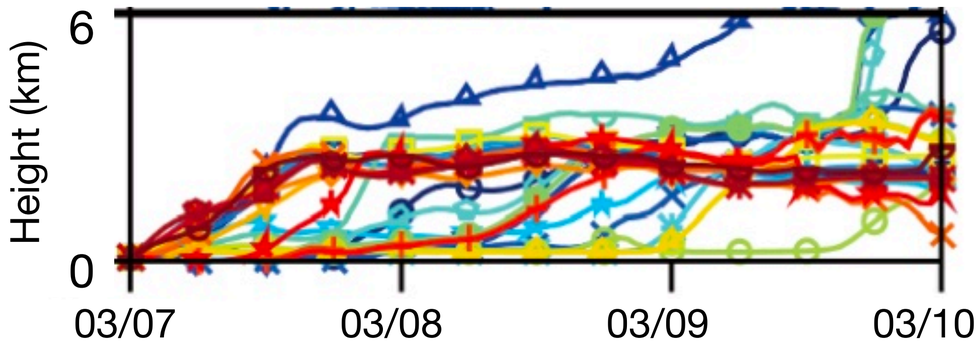


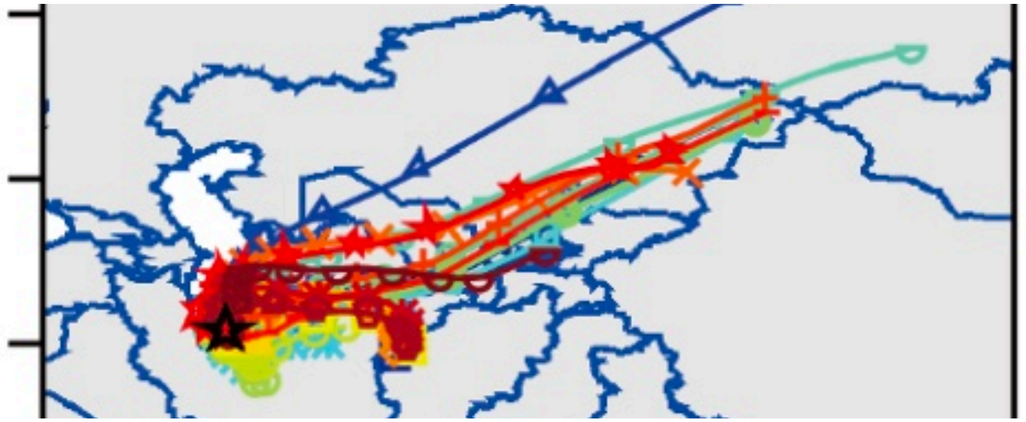
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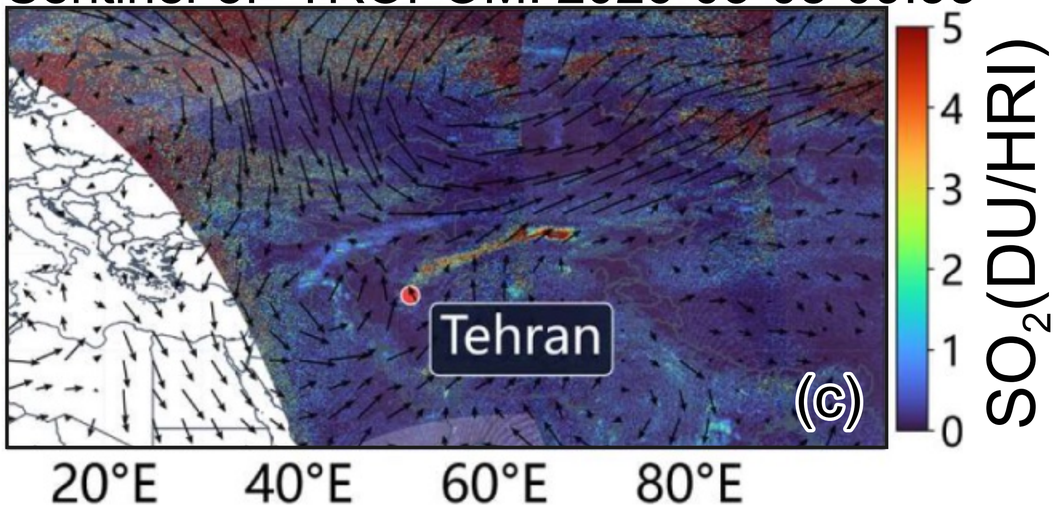


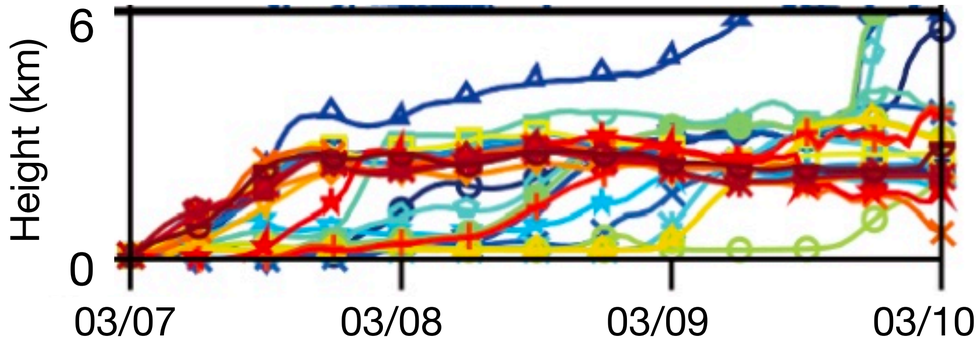
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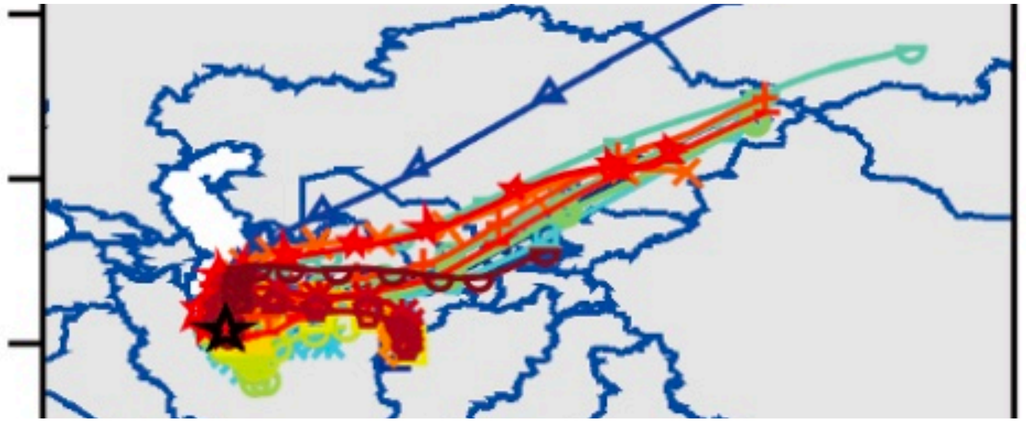
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- The plume has moved **a great distance** but high column densities are still clearly identifiable

Sentinel-5P TROPOMI 2026-03-08 09:55







Take aways

- Aerosols occur in warfare, both as weapons and byproducts of weapon use
- They can have siginificant consequences for the civilian populations, especially in urban warfare
- SO2 emissions can be tracked using satellite measurements and atmospheric transport trajectories help interpret the results
- Emissions can be carried large distances from the original impact site.

References

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